Machine-checked degree lower bounds for interpolating the elliptic curve discrete logarithm from x-coordinates

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Abstract

We record three degree lower bounds for polynomials that interpolate functions of the discrete logarithm from the x-coordinates of multiples kP of a point on a short Weierstrass curve, and we give Lean 4 formalizations of them. (B) If $g(x(kP)) = k^2$ for $1 \le k \le (r-1)/2$, where P has odd order r on a short Weierstrass curve over a field F with $2 \neq 0$ and $3 \neq 0$, then $r \le 6 \deg g + 5$. (C) If $g(x(kP)) = k$ for $1 \le k \le N$ and $2N < r$, then $N \le 3 \deg g + 2$. (D) If the interpolation $g(x(kP)) = k^2$ is required only on a set $A \subseteq [1, N]$, $2N < r$, then for each $j \ge 1$ at most $3 \deg g$ elements $k \in A$ have $k - j, k + j \in A$ (under an explicit extra hypothesis). All three are proved by one device, a cleared second-difference polynomial of exact degree $3 \deg g$ (constant g is treated separately). The statements are verified by Lean against a pinned Mathlib; the Lean statements are quoted verbatim. Interpolation degree bounds of this kind are due to Coppersmith–Shparlinski, Lange–Winterhof and Satoh; we could not access their exact statements, so our bounds may be known or weaker. The contribution claimed is the machine-checked formalization and the precise hypotheses. The results concern polynomial degree in a restricted model and have no bit-complexity consequence. A clearly marked addendum (Section 7) records a formally verified consequence for division-free straight-line programs and a method barrier (a proof attempt, not reviewed).

1 Introduction

The elliptic curve discrete logarithm problem (ECDLP) asks, given P and $Q = kP$ in a group of order r, for k modulo r. A natural question about restricted classes of algorithms is whether the map $x(kP) \mapsto k$ (or a quantity such as k^2, which is the natural x-only invariant because $x(kP) = x(-kP)$) can be given by a low-degree polynomial. Here k is a positive integer representative with $k \le (r-1)/2$, and the target k^2 (or k) is its image in the field F; it is not a function of k modulo r in general (over $\mathbb{F}_p$ with $p \ge 5$ and $k \le (r-1)/2 < p$ no wrap-around occurs, but over extension fields k may exceed the characteristic). A lower bound on the degree of any interpolating polynomial shows that no *low-degree-polynomial formula* computes the logarithm; it is a barrier for one restricted model, in the spirit of the interpolation results cited in Section 6.

What such a bound does *not* say should be stated plainly. It says nothing about algorithms that use other operations (comparisons, branching, group operations, index calculus), nothing about *lower* bounds on the cost of evaluating it, and nothing about bit complexity. Degree and evaluation cost are related only weakly: Horner's rule evaluates a degree-d polynomial with at most d multiplications, so degree is an upper bound for cost in that model; conversely a division-free straight-line program with s multiplications computes a polynomial of degree at most 2^s, so a degree lower bound yields only a *logarithmic* lower bound on the number of multiplications (see Section 7 for the formalized form), and no statement about bit operations. A high-degree polynomial such as X^{p-2} can be cheap to evaluate. The open target of the laboratory (a universal classical ECDLP algorithm with worst-case $O(n \log n)$ bit operations) is unaffected by this report.

Contents. Section 2 fixes notation; Section 3 states the three results; Section 4 gives the proofs; Section 5 quotes the Lean statements and the trust base; Section 6 discusses prior work and novelty; Section 8 lists limitations; Section 7 is a marked addendum of two later laboratory results.

Existence is not at issue: the nodes $x(kP)$, $1 \le k \le N$, $2N < r$, are pairwise distinct (Section 2), so Lagrange interpolation always gives g with $\deg g \le N-1$ for the prescribed values; for Theorem B this is $\deg g \le (r-3)/2$. Theorems B–D therefore determine the minimal interpolation degree up to a constant factor (about 3), and the bounds are linear in r.

2 Setting and notation

Let F be a field and $W : y^2 = x^3 + ax + b$ a short Weierstrass cubic over F ($a_1 = a_2 = a_3 = 0$, $a = a_4$, $b = a_6$). The principal case is $F = \mathbb{F}_p$ and W an elliptic curve with a point P of prime order r. For a nonzero affine point Q we write $x(Q)$ for its x-coordinate, and we write $r = \mathrm{ord}(P)$ for the additive order of P. For a polynomial $g \in F[X]$ we write $m = \deg g$ and $\mathrm{lc}(g)$ for its leading coefficient. Throughout we assume $2 \neq 0$ in F (written $2 \neq 0$; Theorem B also uses $3 \neq 0$).

Two facts are used. First, if $2N < r$ then $k \mapsto x(kP)$ is injective on $\{1, \dots, N\}$: $x(kP) = x(lP)$ forces $kP = \pm lP$, i.e. $k \equiv \pm l \pmod r$, and $2 \le k + l \le 2N < r$ rules out $k \equiv -l$, so $k \equiv l$, hence $k = l$. Second, the x-addition identities: for $Q = (u, v)$, $P = (x_0, y_0)$ affine points with $u \neq x_0$,

$$\big(x(Q+P) + x(Q-P)\big)(u - x_0)^2 = 2\big((x_0 u + a)(u + x_0) + 2b\big), \tag{1}$$

$$x(Q+P)\, x(Q-P)\, (u - x_0)^2 = (x_0 u - a)^2 - 4b(u + x_0). \tag{2}$$

For (1) write $s_\pm = x(Q \pm P) = \big((v \mp y_0)/(u - x_0)\big)^2 - (u + x_0)$ and use $v^2 + y_0^2 = u^3 + x_0^3 + a(u + x_0) + 2b$ together with $(u + x_0)(u - x_0)^2 = u^3 - u^2 x_0 - u x_0^2 + x_0^3$; (2) is a similar computation (it is part of the formalized algebraic core, Section 5).

3 Main results

Theorem B (k^2-interpolation)**.** *Let W be a short Weierstrass curve over a field F with $2 \neq 0$ and $3 \neq 0$ in F, and P a point of odd order r. If $g \in F[X]$ satisfies $g(x(kP)) = k^2$ for $1 \le k \le (r-1)/2$, then $r \le 6 \deg g + 5$; that is, $\deg g \ge (r-5)/6$.*

For an elliptic curve over $\mathbb{F}_p$ and P of prime order $r > 2$ this is the statement recorded as HYP-eda6732a (the ledger statement assumes r prime; the blind review REV-83ec2fc9 notes that "r prime" can be weakened to "r odd", which is the formalized form). The hypothesis $3 \neq 0$ is formalized but, according to REV-15104e72, not needed mathematically (see Section 4).

Theorem C (linear interpolation)**.** *Let W be short Weierstrass over F with $2 \neq 0$, P a point with $\mathrm{ord}(P) = r$, and $2N < r$. If $g \in F[X]$ satisfies $g(x(kP)) = k$ in F for $1 \le k \le N$, then $N \le 3 \deg g + 2$. In particular, for $N = (r-1)/2$ (r odd), $r \le 6 \deg g + 5$.*

(HYP-42f91290.) The assumption $3 \neq 0$ is not needed.

Theorem D (partial agreement)**.** *Let W be short Weierstrass over F with $2 \neq 0$, $\mathrm{ord}(P) = r$, $2N < r$ (for r odd, equivalently $N \le (r-1)/2$), $A \subseteq \{1, \dots, N\}$, and $g \in F[X]$ with $g(x(kP)) = k^2$ for all $k \in A$. For $j \ge 1$ let $S_j = \#\{k \in A : k - j \in A,\ k + j \in A\}$ (the k with $j < k$ automatically). If $\deg g \ge 1$ or $j \neq 0$ in F, then $S_j \le 3 \deg g$.*

(HYP-a1dd5501, part (2); the main inequality only.) The side condition is needed over general fields. An instance found by the adversarial review REV-15104e72 (checked there by

computer, not by us and not in Lean): over $\mathbb{F}_{27} = \mathbb{F}_3[t]/(t^3 - t + 1)$ the curve $y^2 = x^3 + t^2 x$ has a point P of order 28; with $N = 7$ (so $2N = 14 < 28$), $A = \{1, 4, 7\}$, $g = 1$ and $j = 3$ one has $g(x(kP)) = k^2$ on A ($k^2 \in \{1, 16, 49\} = \{1\}$ in characteristic 3), $\deg g = 0$ and $S_3 = 1 > 0$. This instance needs $r > 4p + 2$, which is impossible over $\mathbb{F}_p$ by the Hasse argument below; that argument does not extend to $\mathbb{F}_q$. Over $\mathbb{F}_p$ the condition excludes only constant g with $p \mid j$, and then $S_j = 0$ for the following reason (*not* formalized in Lean): $S_j \geq 1$ requires $k - j \geq 1$ and $k + j \leq N$, so $N \geq 2j + 1 \geq 2p + 1$, whence $r > 2N \geq 4p + 2$, contradicting the Hasse bound $r \leq \#E(\mathbb{F}_p) \leq p + 1 + 2\sqrt{p}$. Hence over $\mathbb{F}_p$ the unconditional informal statement $S_j \leq 3 \deg g$ holds. (One must not argue $j \leq N < p$; that is false in general, e.g. $y^2 = x^3 + 2x + 1$ over $\mathbb{F}_3$ with $r = 7$, $N = 3$.)

Remark 1 (Hypotheses and the refuted transcription). The hypothesis $2N < r$ in Theorem D cannot be dropped. The ledger record HYP-5da6025e (status: refuted) contains a transcription of part (2) with $N < \min(p, r)$ in place of $N \leq (r-1)/2$; the blind review REV-ac47cc67 produced the counterexample $y^2 = x^3 + 5$ over $\mathbb{F}_7$, $P = (3, 2)$ of order $7 = p$, $A = \{1, \ldots, 6\}$, $g = 5X$, which satisfies $g(x(kP)) = k^2$ for $k = 1, \ldots, 6$ with $\deg g = 1 < S_1/3 = 4/3$. The hypothesis $N \leq (r-1)/2$ had been present in the original agent report and was lost in a transcription error by the principal investigator; the corrected statement is HYP-a1dd5501 and is what is formalized here. Ledger parts of HYP-a1dd5501 not covered here: part (1) for rational functions on E (reviewed, not formalized), and the density corollary $\deg g \geq ((3\delta - 2)N - 2)/3$ for $\delta > 2/3$ (reviewed, not formalized).

Remark 2 (Background: Theorem A). The ledger also records (HYP-adb581ff, status *open*, reviewed, not formalized) a result for rational functions f on E with $f(kP) = k$ for $1 \leq k \leq N \leq r - 1$: the pole-divisor degree is at least $(N-1)/2$, and polynomial total degree at least $(N-1)/6$. It is recorded as known in kind (Section 6) and is cited only as background.

4 Proofs

Assume $m = \deg g \geq 1$ for the moment; constant g is not covered by the exact-degree argument and is treated separately at the end of each case. Fix an affine point $P_0 = (x_0, y_0)$ ($P_0 = P$ in Theorems B, C; $P_0 = jP$ in Theorem D) and put $d = u - x_0$.

The cleared second-difference polynomial. For a variable u let $s_\pm$ be the (formal) roots of $z^2 - e_1 z + e_2$ where $e_1 = s_+ + s_-$ and $e_2 = s_+ s_-$ are the rational functions in u given by (1) and (2) (so $g(s_+) + g(s_-)$ is *defined* as the symmetric expression $\sum_i c_i p_i(e_1, e_2)$ below, and is evaluated at $u = x(Q)$ only after clearing denominators): $e_1 = n_1/d^2$, $e_2 = n_2/d^2$ with n_1, n_2 polynomials of degree ≤ 2. Write $g = \sum_{i \leq m} c_i X^i$. Then $g(s_+) + g(s_-) = \sum_i c_i p_i$ with $p_i = s_+^i + s_-^i$ a polynomial with integer coefficients in e_1, e_2 whose monomials $e_1^\alpha e_2^\beta$ satisfy $\alpha + 2\beta = i$ (Newton–Girard). Multiplying a monomial by d^{2m} gives $n_1^\alpha n_2^\beta d^{2(m-\alpha-\beta)}$, a polynomial of degree $\leq 2(\alpha + \beta) + 2(m - \alpha - \beta) = 2m$, because $\alpha + \beta \leq i \leq m$. Define

$$G_c(u) = d^{2m}\big[g(s_+) + g(s_-)\big] - 2d^{2m} g(u) - c\, d^{2m} \in F[u],$$

for a constant c. The first term has degree $\leq 2m$, the third degree $\leq 2m$ (and 0 when $c = 0$), and $-2d^{2m} g(u)$ has degree exactly $3m$ with leading coefficient $-2\,\mathrm{lc}(g) \neq 0$ (here $2 \neq 0$ is used). Since $2m < 3m$ for $m \geq 1$:

$$\deg G_c = 3m \quad \text{for every constant } c, \quad \text{in particular } G_c \neq 0. \tag{3}$$

Roots. Suppose $Q = kP$ is affine with $x(Q) \neq x_0$ and $Q \pm P_0$ affine. Then evaluating at $u = x(Q)$, the quantities $s_\pm$ are $x(Q \pm P_0)$ by the identities of Section 2, so $G_c(x(Q)) = d^{2m}\,[g(x(Q + P_0)) + g(x(Q - P_0)) - 2g(x(Q)) - c]$ with $d \neq 0$.

Theorem B. Take $P_0 = P$, $c = 2$. For $2 \le k \le (r-3)/2$ we have $1 \le k-1$ and $k+1 \le (r-1)/2$, so all three values of g are prescribed and the bracket equals $(k+1)^2 + (k-1)^2 - 2k^2 - 2 = 0$. The values $x(kP)$, $2 \le k \le (r-3)/2$, are pairwise distinct (injectivity) and differ from $x(P)$ (since $k \ge 2$), so G_2 has at least $(r-5)/2$ distinct roots. By (3), $(r-5)/2 \le 3m$, i.e. $r \le 6m+5$. Constant $g = c_0$: if $r \le 5$ the bound is trivial; if $r \ge 7$ then $(r-1)/2 \ge 3$ and $g(x(P)) = 1$, $g(x(2P)) = 4$ force $3 = 0$, which $3 \ne 0$ excludes. (According to REV-15104e72, $3 \ne 0$ is superfluous: in characteristic 3 and $r \ge 7$, $k = 3$ gives $9 = 0 \ne 1 = g$. We keep it because it is the formalized hypothesis.) Hence the bound holds.

Theorem C. Take $P_0 = P$, $c = 0$. For $2 \le k \le N-1$ the bracket is $(k+1)+(k-1)-2k = 0$, giving $N-2$ distinct roots (injectivity as $2N < r$), so $N - 2 \le 3m$. If g is constant and $N \ge 2$, then $g(x(P)) = 1 \ne 2 = g(x(2P))$, contradiction; so $N \le 1 \le 3m + 2$.

Theorem D. Take $P_0 = jP$ (note $j < k \le N < r$). For $k \in C_j = \{k \in A : j < k,\ k \pm j \in A\}$ the bracket with $c = 2j^2$ is $(k+j)^2 + (k-j)^2 - 2k^2 - 2j^2 = 0$. The values $x(kP)$ are distinct for distinct $k \in C_j$ and differ from $x(jP)$ because $k \ne j$ and $k + j \le 2N < r$, so $|C_j| \le 3m$ by (3). If $g = c_0$ is constant, the bracket is $-2j^2 \ne 0$ whenever $j \ne 0$ in F, so G has no roots and $C_j = \emptyset$. □

Exact degree. The computation above shows that, for the second-difference identity, the cleared polynomial has degree exactly $3 \deg g$ and leading coefficient $-2\,\mathrm{lc}(g)$; for Theorem C this is recorded in HYP-42f91290 and agrees with the numerical checks of the ledger.

5 Formal verification

Environment. Toolchain `leanprover/lean4:v4.35.0` (file `lean-toolchain`); Mathlib revision `56a005855ea17e0b0958f73ec801ae140b0a1679` (`lakefile.toml`, with `-DwarningAsError=true`). Source commits recorded in the proof records: Theorem B, PRF-34538f75, commit `cc8dc791a0bd74f89b6cd614898a5773abdc500a` (`lake build`: 3163 jobs, 0 warnings); Theorem C, PRF-35d1387a, commit `8191c0f69398b632fc2c7c7efbfdcdb9e2634b7b` (3164 jobs, 0 warnings); Theorem D, PRF-5357b698, commit `84ddbad82b2f1412e851cf1ab6012b6650bba544` (3165 jobs, 0 warnings). Files: `MathResearch/ECDLP/Interpolation.lean` (generic polynomial core), `CurveX.lean`, `LinearInterp.lean`, `PartialAgreement.lean`. All three proofs use Mathlib's `WeierstrassCurve.Affine.Point`; the curve-specific input is the x-addition identities (`CurveX.sum_prod_identity`, `CurveX.smul_relations`) and injectivity of $k \mapsto x(kP)$.

Statements (verbatim from the repository). Here `xCoord` is the x-coordinate function on `W.Point`, defined to be 0 at the point at infinity (harmless here, since every point evaluated is kP with $1 \le k < r$, which is affine). The binders `{F : Type*} [Field F] [DecidableEq F]` come from `variable` lines of the files; `Affine` is `WeierstrassCurve.Affine` (via `open WeierstrassCurve`) and `F[X]` is `Polynomial F`.

The declarations live in the Lean namespace `MathResearch.ECDLP.CurveX`, in these files: Theorem B in `CurveX.lean`, Theorem C in `LinearInterp.lean`, Theorem D in `PartialAgreement.lean`.

```
theorem degree_bound_short_weierstrass {W : Affine F} (h1 : W.a₁ = 0) (h2 : W.a₂ = 0)
    (h3 : W.a₃ = 0) (h2' : (2 : F) ≠ 0) (h3' : (3 : F) ≠ 0) (P : W.Point) (r : ℕ)
    (hr : addOrderOf P = r) (hodd : r % 2 = 1) (g : F[X])
    (hg : ∀ k : ℕ, 1 ≤ k → k ≤ (r - 1) / 2 → g.eval (xCoord (k • P)) = (k : F) ^ 2)
     :
    r ≤ 6 * g.natDegree + 5
```

```
theorem degree_bound_linear_short_weierstrass {W : Affine F} (h1 : W.a₁ = 0) (h2 : W.a₂
    = 0)
    (h3 : W.a₃ = 0) (h2' : (2 : F) ≠ 0) (P : W.Point) (r : ℕ)
    (hr : addOrderOf P = r) (N : ℕ) (hN : 2 * N < r) (g : Polynomial F)
    (hg : ∀ k : ℕ, 1 ≤ k → k ≤ N → g.eval (xCoord (k • P)) = (k : F)) :
    N ≤ 3 * g.natDegree + 2
```

```
theorem partial_agreement_degree_bound {W : Affine F} (h1 : W.a₁ = 0) (h2 : W.a₂ = 0)
    (h3 : W.a₃ = 0) (h2' : (2 : F) ≠ 0) (P : W.Point) (r : ℕ) (hr : addOrderOf P = r)
    (N : ℕ) (hN : 2 * N < r) (A : Finset ℕ) (hA : ∀ k ∈ A, 1 ≤ k ∧ k ≤ N) (g : F[X])
    (hg : ∀ k ∈ A, g.eval (xCoord (k • P)) = (k : F) ^ 2) (j : ℕ) (hj : 1 ≤ j)
    (hjg : 1 ≤ g.natDegree ∨ (j : F) ≠ 0) :
    (A.filter (fun k => j < k ∧ k - j ∈ A ∧ k + j ∈ A)).card ≤ 3 * g.natDegree
```

The full names are `MathResearch.ECDLP.CurveX.degree_bound_short_weierstrass`, `MathResearch.ECDLP.CurveX.degree_bound_linear_short_weierstrass` and `MathResearch.ECDLP.CurveX.partial_agreement_degree_bound`.

Correspondence to the informal statements. (i) The formal statements are for an arbitrary field F, not only $\mathbb{F}_p$, and for any point P of the given additive order; r need not be prime. Theorem B needs r odd, $2 \neq 0$ and $3 \neq 0$; Theorems C and D need neither $3 \neq 0$ nor r odd (the condition $2N < r$ is equivalent to $N \leq (r-1)/2$ for r odd). (ii) Since `W : Affine F` is not assumed nonsingular, the review REV-1f44f52b notes that the results also cover nonsingular points of singular cubics. (iii) Theorem D carries the extra hypothesis `hjg`; the Hasse-bound argument that removes it over $\mathbb{F}_p$ (Section 3) is *not* formalized. (iv) The formal statements are degree bounds only; they assert no existence of g and no complexity. REV-1f44f52b and REV-6c9bfbc1 accepted the correspondence of B and C; the independent review REV-685408b5 (formal correctness only; no complexity or novelty verification) verified the correspondence for D.

Trust base. `#print axioms` on each of the three declarations reports exactly `[propext, Classical.choice, Quot.sound]`. The evidence is per theorem (we verified that each file contains the corresponding `#print axioms` line): Theorem B, `research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-b585f279-fd16-4cca-b3a1-c2484d545078.evidence` (`degree_bound_short_weierstrass`, together with `sum_prod_identity` and `smul_relations`); Theorem C, `research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-5da55f17-bbac-4f82-9271-da5f30d66b80.evidence` (`degree_bound_linear_short_weierstrass`); Theorem D, `research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-ca0a1a26-2ee3-4da0-be6c-96a56ffa6cc6.evidence` (`partial_agreement_degree_bound`). Trust therefore rests on the Lean kernel, Mathlib at the pinned revision, and the faithfulness of the statements to the informal ones (reviewed by AI agents of the same provider; independence is partial).

Reproduction. At a checkout of the repository: `lake build` (pinned toolchain and Mathlib), followed by `#print axioms MathResearch.ECDLP.CurveX.degree_bound_short_weierstrass` and likewise for the other two names (see `research/ecdlp/proofs/checks/Round8Axioms.lean` for the pattern).

Empirical sanity checks (not part of the proof). The ledger records toy-size numerical checks: the x-addition identity 2519 times with no failure (Theorem B record); 208 toy cases without violation (REV-83ec2fc9); for Theorem C 533 curves with $p \leq 211$, then 12,459 cases including extension fields (REV-6c9bfbc1); 1242 toy points for B (REV-1f44f52b); 739,847 pairs

(A, j) for D (a weak check). These tests do not constitute proofs and are not needed given the Lean proofs; they indicate the bounds are far from tight in tests.

6 Relation to prior work and novelty status

Coppersmith and Shparlinski [1] studied polynomial approximation of the discrete logarithm and the Diffie–Hellman mapping (title of [1]); we did not access the text. Lange and Winterhof (COCOON 2002 [2]; AAECC-15 2003 [3]) and Satoh (WCC 2005, published 2006 [4]) prove degree lower bounds for polynomial interpolation of the elliptic curve discrete logarithm and related maps. **The exact statements of these results were not accessible to the laboratory**: only bibliographic records and abstracts were seen, and full texts were not accessed. We therefore cannot compare constants or hypotheses. Our bounds may be known or weaker. Theorem A is recorded in the ledger as known in kind; the laboratory's novelty checks (ECDLP-22) found no result of the form of Theorems B–D in the accessible material, but this is absence of evidence, not a novelty claim. The laboratory's ledger record for Theorem C (HYP-42f91290) says it is "likely implied by / weaker than Lange–Winterhof-type results" and is *not* claimed novel. Novelty of all three is *unsettled*.

Claimed contribution: a machine-checked formalization (Lean 4, Mathlib, standard axioms only) of the three degree bounds, with explicit, reviewed hypotheses (including the corrected hypothesis $2N < r$ of Theorem D and its generality over arbitrary fields). Bibliographic metadata (first names, venue, pages, year, DOI) of the cited entries were checked against Crossref records (by REV-2b383b9b for [2], and in this revision for [3], [4] and [1]); in particular the year of [4] is 2006. Full texts were still not accessed.

7 Addendum: later laboratory results (not part of Theorems B–D)

This section was added in revision 1 and is clearly separate from the results above. Its two items carry the statuses stated here.

(i) Division-free straight-line programs: formally verified. Ledger record PRF-58cf340e (derived from HYP-92a9b4d9; status *formally verified*; independent review pending, not yet reviewed). In `MathResearch/ECDLP/SLPBound.lean` a *program* over a commutative ring F is a list of operations with a single input variable x; each operation appends a constant of F (free), the input, or the sum, difference or product of two earlier values. There are no divisions, comparisons or branching. A program of length L computes a polynomial of degree at most 2^L in x. Combined with Theorem C this gives, in Lean (source commit `bff5e14c0f187719fb79c07f7768a2856f20604a`, `lake build` 3166 jobs, 0 warnings; build evidence `research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-94bd6bd6-9c44-47e7-9743-cd567e803d9f.evidence`; axioms `[propext, Classical.choice, Quot.sound]` in `research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-aeddfabd-5935-4ad5-9911-2c48a87063dc.evidence`):

```
theorem length_bound_linear_short_weierstrass {W : Affine F} (h1 : W.a₁ = 0) (h2 : W.a₂
    = 0)
    (h3 : W.a₃ = 0) (h2' : (2 : F) ≠ 0) (P : W.Point) (r : ℕ) (hr : addOrderOf P = r)
    (N : ℕ) (hN : 2 * N < r) (prog : List (Op F)) (i : ℕ)
    (hprog : ∀ k : ℕ, 1 ≤ k → k ≤ N → (evalAt prog (xCoord (k • P))).getD i 0 = (k
    : F)) :
    N ≤ 3 * 2 ^ prog.length + 2
```

(namespace `MathResearch.ECDLP.SLPBound`; `evalAt` evaluates the program on the input, and out-of-range indices read 0.) Thus a program of length L that outputs k at $x(kP)$ for $1 \le k \le N$, $2N < r$, needs $L \ge \log_2((N-2)/3)$. This answers directly the distinction between polynomial degree and program length raised in review REV-15104e72 (N1): the bound is on the number of ring operations in this restricted model, logarithmic in N. The cost-model consequence recorded in the ledger (if each field operation costs $\Omega(\log p)$ bit operations, then $\Omega(n^2)$ bit operations are needed in this model) is a remark, *not formalized*, and concerns only this restricted model; it says nothing about the open $O(n \log n)$ target for general algorithms.

(ii) Method barrier L2: proof attempt, PI-checked, not reviewed. Ledger record HYP-0d4f14da (status *open*, proof status *proof attempt*; the principal investigator checked the proof steps and reproduced the supporting numerical scripts; not formalized, not independently reviewed). The claim: for a "translation annihilator" $D = \sum_{i \in S} c_i G(Q+iP)$ with $\sum c_i = \sum i c_i = 0$ ($|S| = s$, width w), $G = g \circ x$, $\deg g = d \ge 1$, $2N < r$, one has $N \le ds + w$, minimized at $s = 3$, $w = 2$, which is Theorem C. Hence zero-counting arguments with translates cannot improve the constant $1/3$; any sharper bound must use information beyond the pole structure and the values $G(kP) = f(k)$. We state this only as recorded; it has no proof in this report.

(iii) Empirical conjecture (not proved). Ledger record HYP-a8d1f541 ("Conjecture C*", status *open*, experimental): outside structured families (for example anomalous CM $j = 0$ curves), the minimal degree of an interpolating g is $N - 1 - O(1)$. The ledger records 154,288 toy cases over prime fields with $p \le 997$ and a maximal observed deficit of 3. This is an observation at toy sizes, not evidence for large parameters. If true it would show that the factor 3 between the bounds above and the trivial Lagrange upper bound is a limitation of the method, not of the problem.

8 Limitations

- Degree is not time. A lower bound on $\deg g$ for polynomials in one variable $x(kP)$ does not bound the running time of any algorithm, and we draw no bit-complexity consequence.
- The results concern interpolation from x-coordinates only, with exact agreement on a range (B, C) or a subset (D); they say nothing about approximate or probabilistic computation or about other coordinates.
- Existence and tightness: interpolating polynomials of degree $\le N - 1$ exist by Lagrange interpolation, so the bounds are tight up to a constant factor; we prove no exact value of the minimal degree (see the conjectural remark in Section 7).
- The ledger's rational-function versions (Theorem A, part (1) of the partial-agreement hypothesis) and the density corollary are not formalized.
- Theorem D's extra hypothesis is removed over $\mathbb{F}_p$ only by an informal Hasse-bound argument.
- The straight-line-program bound of Section 7 is for a restricted, division-free, single-input model; its bit-cost corollary is not formalized, and the method-barrier and conjecture items there are unreviewed and unproved respectively.
- The universal $O(n \log n)$ ECDLP target remains open; nothing here improves on any known ECDLP complexity.
- Novelty is unsettled (Section 6); no human has verified the mathematics, the Lean-to-statement correspondence, or the bibliography.

Provenance note. This report was produced by an AI research laboratory (this draft and its revision 1 by worker agents reporting model identity Sonnet 5.5, `claude-sonnet-5-5`, not verifiable from runtime metadata) and reviewed by AI agents; no human verification has yet taken place. It is an internal technical report and not a submission.

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